

Analog CMOS IC Design (EE651)

Assignment- 3

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1. Design Common Source (CS) amplifiers with active load for the following specifications.

Voltage Gain > 100 V/V, **Power dissipation** $< 10 \mu\text{W}$, min input voltage swing = $100 \mu\text{V}$, max input voltage swing = 12mV .

- Calculate the output swing voltage and gain for the minimum and maximum input swings using transient analysis, and also plot the transient response.
- Plot frequency response (both Gain and Phase response) of the amplifier. Extract Bandwidth and Unity Gain Frequency (UGF) and explain the results.

Fig:-1 schematic of common source amplifier with active load (Component Parameters)

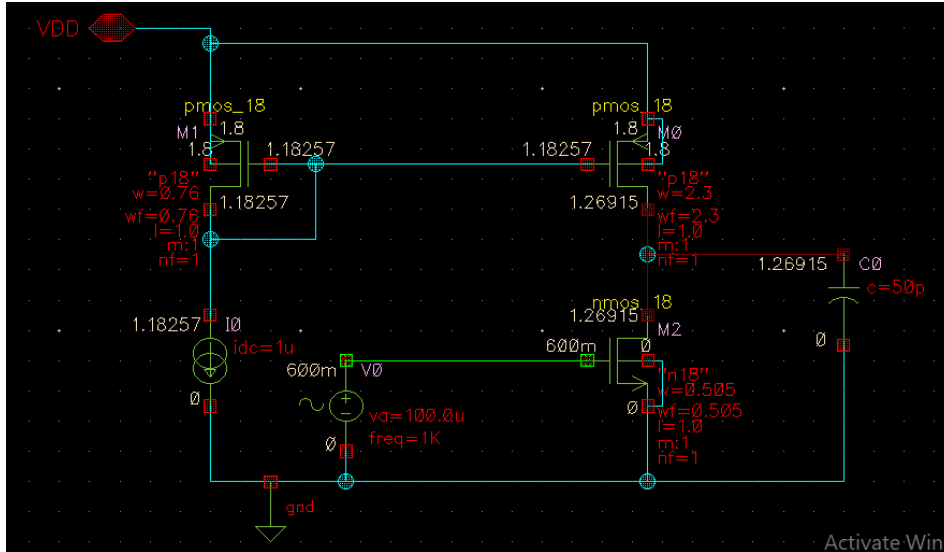


Fig:-2 schematic of common source amplifier with active load(with DC operating points)

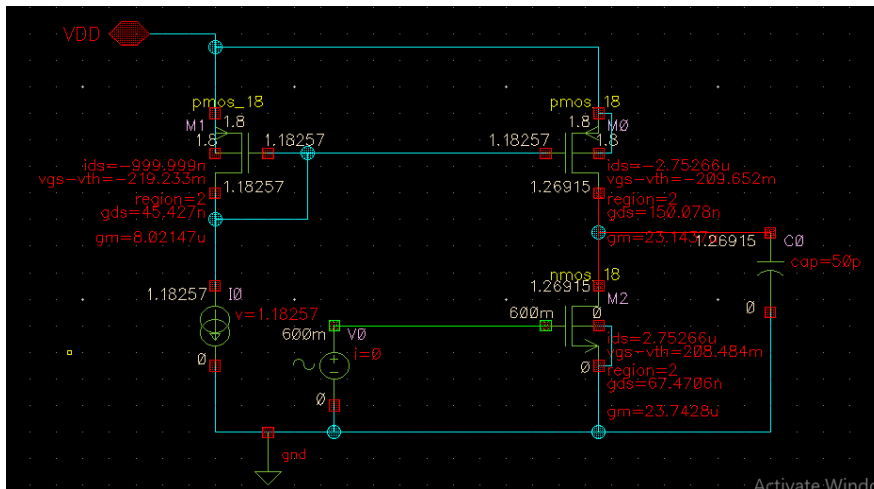


Fig:-3 ADL analysis(common source amplifier)

Outputs				
Name/Signal/Expr	Value	Plot	Save	Save Options
1 Vout		☒	☐	allv
2 Vin		☒	☐	allv
3 Gain Wave	wave	☒	☐	
4 Gain Mag(dB)	40.7593	☒	☐	
5 BandWidth	690.607	☒	☐	
6 UGB	76.2017K	☒	☐	
7 Power	5.57222u	☒	☐	

Bandwidth = 690 Hz

Gain of the CS Amp = 40.76 dB (109.14 V/V)

UGB= 76.2 kHz

The power consumption is 5.57 uW

Fig:-4 common source amplifier with active load(Gain and Swing)

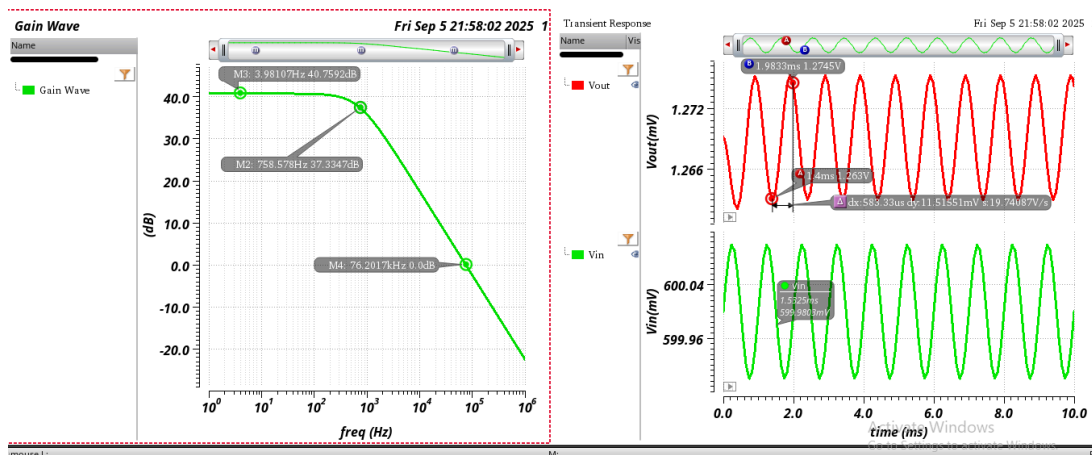
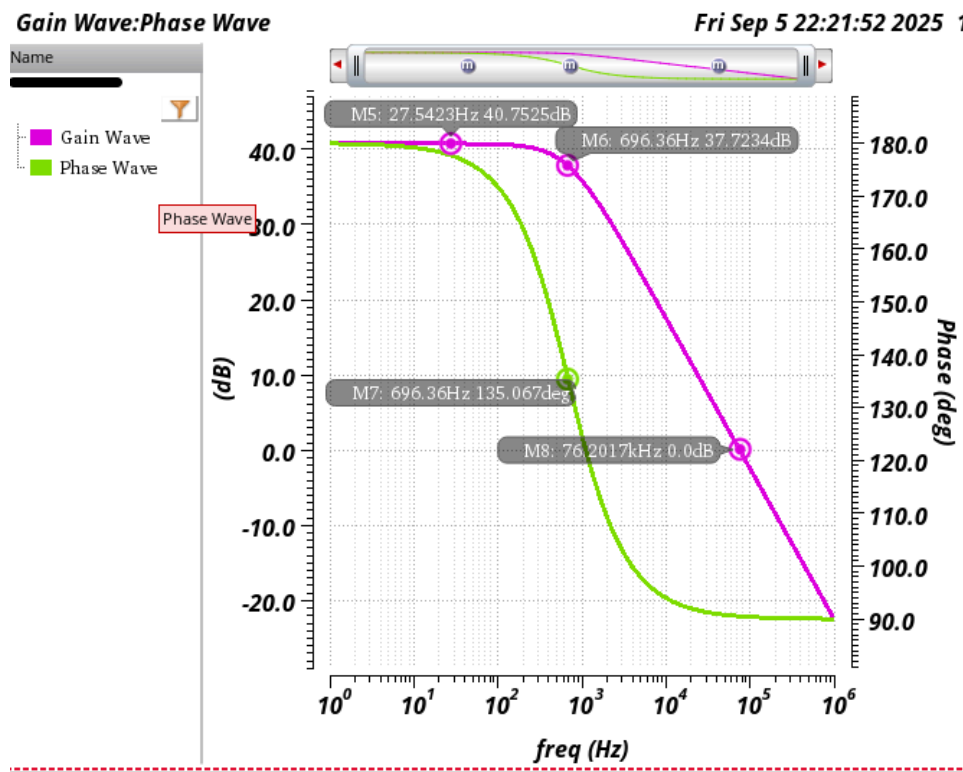


Fig:-5 common source amplifier with active load(Gain and phase plot)



2. Design a Common Drain (CD) amplifier with active load for the following specification.

Voltage gain > 0.95 V/V, **Power dissipation** < 2 mW, **Output impedance (Z_{out})** < 100 Ω . min input voltage swing = 20 mV, max input voltage swing = 1 V.

- Calculate the output swing voltage and gain for the minimum and maximum input swings using transient analysis, and also plot the transient response.
- Plot frequency response (both Gain and Phase response) of the amplifier. Extract Bandwidth and Unity Gain Frequency (UGF) and explain the results.

Fig:-6 schematic of common source amplifier with active load(with DC operating points)

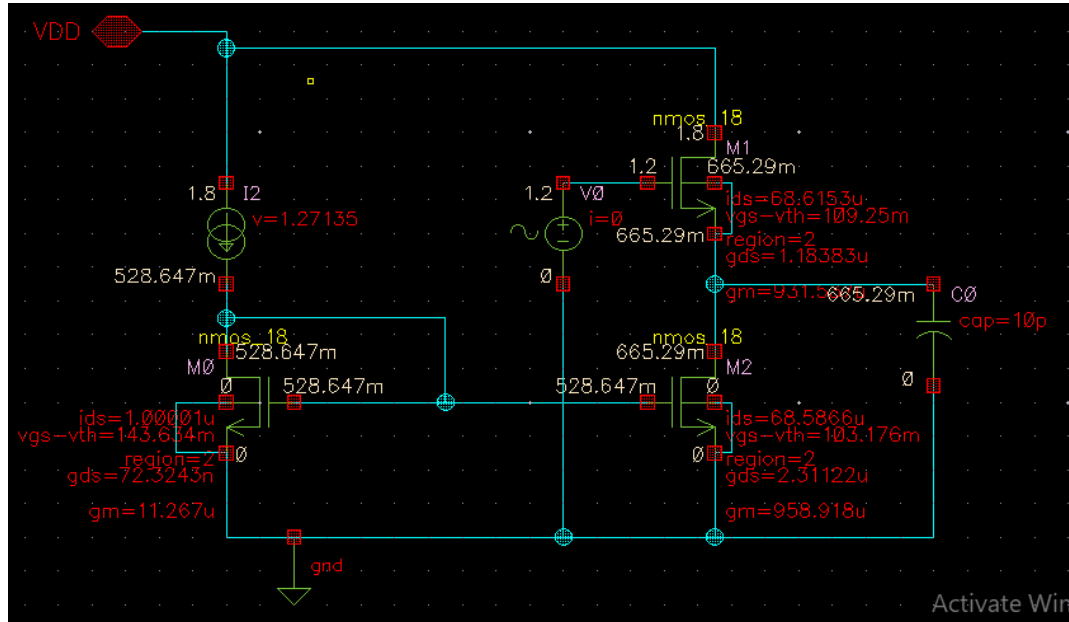


Fig:-7 schematic of common drain amplifier with active load(with component parameters)

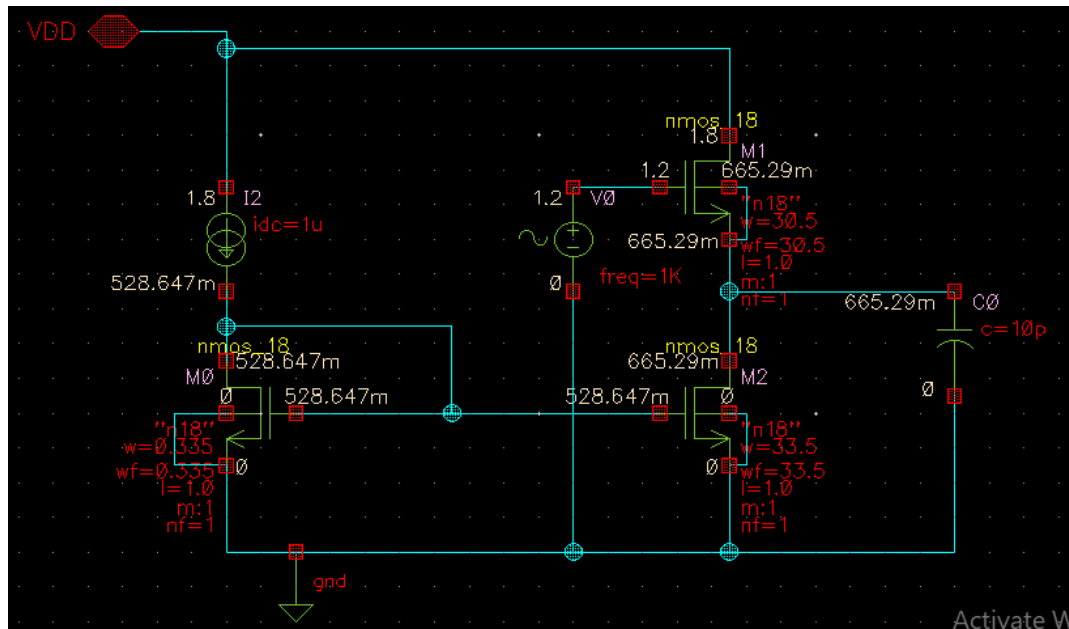
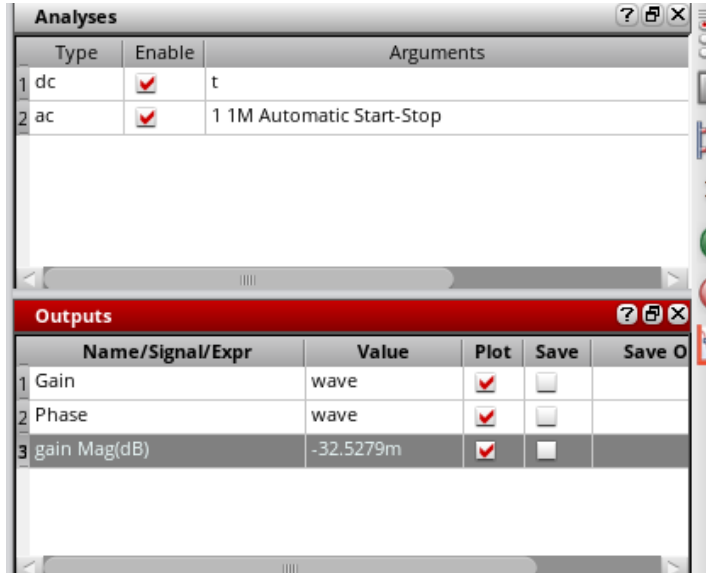


Fig:-8 schematic of common drain amplifier(ADL)



Power consumption =125.298uW

Gain in dB= -32.5dB (0.996 V/V)

Fig:-9 common drain amplifier with active load(Gain and phase plot)

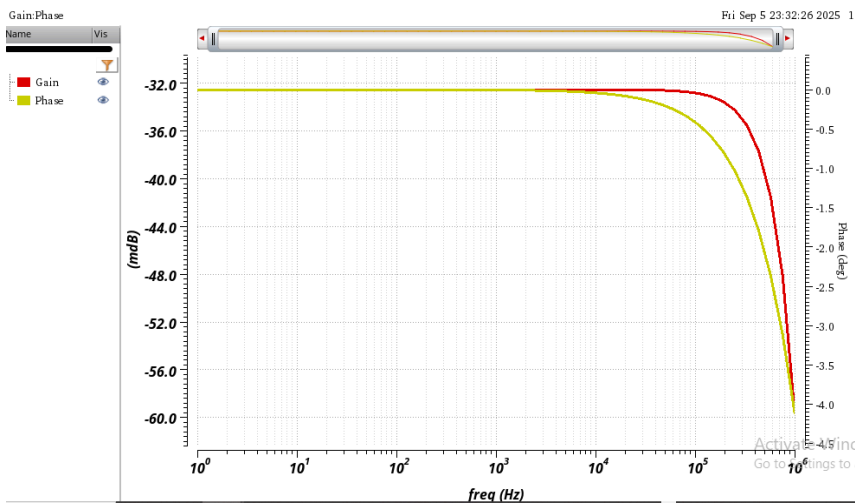


Fig:-10 common drain amplifier with active load output Swing

